CV of Plastic and Immutable Scenario with different S.D.

```
pacman::p_load(ggplot2,#plots
               tictoc, #timing
               dplyr, #filter
               e1071 #skewness,
)
```

```
source("../functions_main/other_functions.R")
```

Space

Space

Space

CV analysis

I computed separately on the server 500 simulation for each case:

Case	Trait given	Coefficient Trait	Type	Standard deviation Trait
1	Once	1	None	0
2	Once	1	Small	0.15
3	Once	1	Moderate	0.3
4	Once	1	High	0.45

Case	Trait given	Coefficient Trait	Type	Standard deviation Trait
5	Every year	1	None	0
6	Every year	1	Small	0.15
7	Every year	1	Moderate	0.3
8	Every year	1	High	0.45

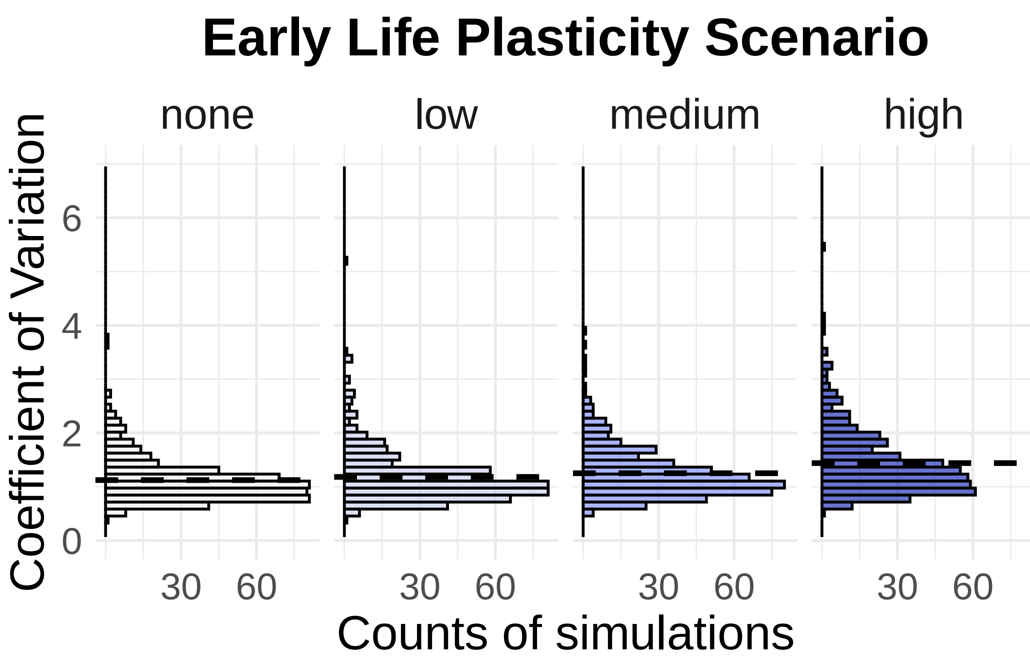
Immutable scenario

```
case1 <-readRDS("../server_runs/RDS/case1.rds")
case2<-readRDS("../server_runs/RDS/case2.rds")
case3<- readRDS("../server_runs/RDS/case3.rds")
case4 <- readRDS("../server_runs/RDS/case4.rds")
```

```
immutable_cv <- bind_rows(
  data.frame(cv = coefficient_of_variation(case1), sd = "none"),
  data.frame(cv = coefficient_of_variation(case2), sd = "low"),
  data.frame(cv = coefficient_of_variation(case3), sd = "medium"),
  data.frame(cv = coefficient_of_variation(case4), sd = "high")) |>
  mutate(sd = factor(sd, levels = c("none","low", "medium", "high")))
```

```
mycolors <- c("#edeef5","#c5caf0","#7083fa", "#0317ad")
p1 <- ggplot(immutable_cv, aes(x = cv, fill = sd)) +
  geom_histogram(binwidth = 0.13, color = "black", alpha = 0.6) +
  geom_vline(
    data = immutable_cv |> group_by(sd) |> summarise(mean_cv = mean(cv)),
    aes(xintercept = mean_cv),
    color = "black", linetype = "dashed", linewidth = 1
  ) +
  labs(x = "Coefficient of Variation", y = "Counts of simulations") +
  theme_minimal() +
  coord_flip() +
  xlim(0,7)+
  scale_y_continuous(breaks = c(30, 60))+
  facet_wrap(~sd, ncol = 4, strip.position = "top") +
  scale_fill_manual(values = mycolors) +
  ggtitle("Early Life Plasticity Scenario") +
  theme(
    legend.position = "none",
    axis.title.x = element_text(size = 18),
    axis.title.y = element_text(size = 18),
    axis.text.x = element_text(size = 14),   # numeri asse x più grandi
    axis.text.y = element_text(size = 14),   # numeri asse y più grandi
    plot.title = element_text(size = 20, face = "bold", hjust=0.5),
    strip.text = element_text(size = 16)
  )
p1
```

```
Warning: Removed 8 rows containing missing values or values outside the scale range
(`geom_bar()`).
```




```r
ggsave("plot1.png", plot = p1, width = 5, height = 6, dpi = 300)
```

```
Warning: Removed 8 rows containing missing values or values outside the scale range
(`geom_bar()`).
```

```r
immutable_cv_table<- immutable_cv |>
       group_by(sd) |>
            summarise(mean_cv = mean(cv))
immutable_cv_table
```

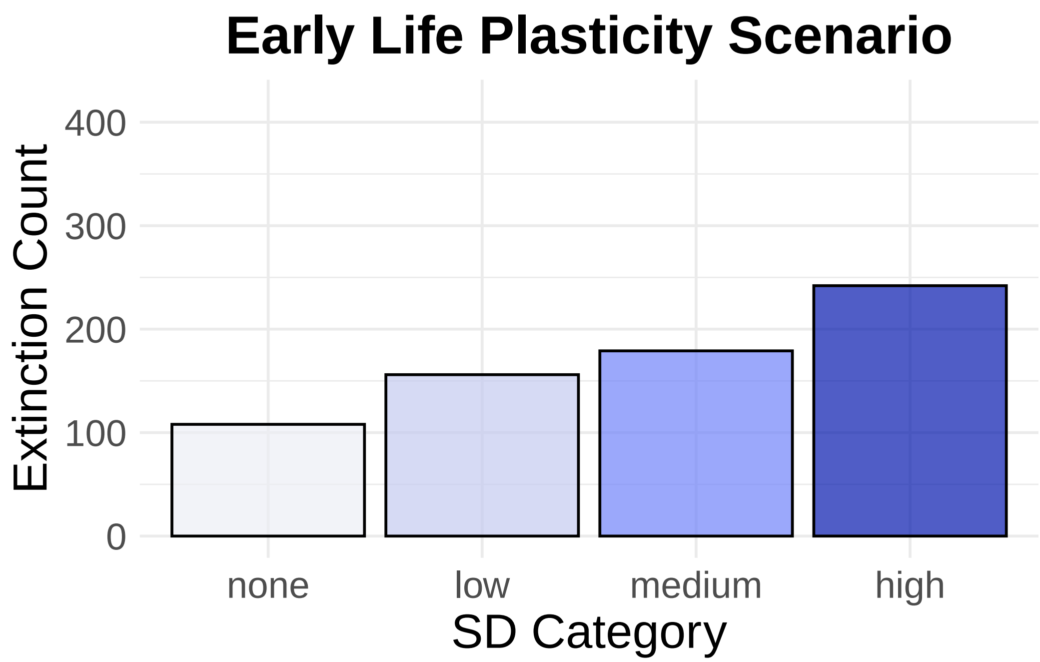
```
# A tibble: 4 x 2
  sd      mean_cv
  <fct>     <dbl>
1 none       1.12
2 low        1.18
3 medium     1.25
4 high       1.44
```

```r
immutable_extinction <- c(extinct_count(case1),extinct_count(case2),extinct_count(case3), ex
sd_labels <- c("none","low", "medium", "high")

immutable_extinction_df <- data.frame(
  count = immutable_extinction,
  sd = factor(sd_labels, levels = sd_labels)
)
```

```r
p2 <- ggplot(immutable_extinction_df, aes(x = sd, y = count, fill = sd)) +
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  scale_fill_manual(values = mycolors) +
  labs(
    x = "SD Category",
    y = "Extinction Count",
    title = "Early Life Plasticity Scenario"
  ) +
  ylim(0,420)+
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.title.x = element_text(size = 18),
    axis.title.y = element_text(size = 18),
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14),
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5)
  )

p2
```




```
ggsave("plot2.png", plot = p2, width = 5, height = 6, dpi = 300)
```

Plasticity scenario

Every year the traits of all the alive animals are updated

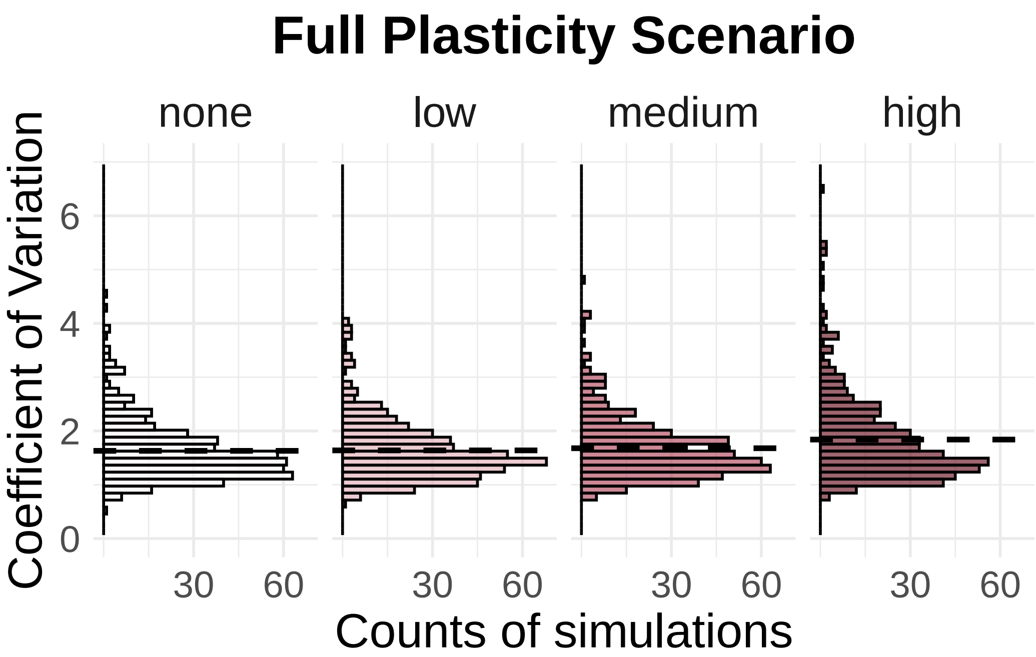
```
case5 <-readRDS("../server_runs/RDS/case5.rds")
case6<-readRDS("../server_runs/RDS/case6.rds")
case7<- readRDS("../server_runs/RDS/case7.rds")
case8 <- readRDS("../server_runs/RDS/case8.rds")
```

```
plasticity_cv <- bind_rows(
  data.frame(cv = coefficient_of_variation(case5), sd = "none"),
  data.frame(cv = coefficient_of_variation(case6), sd = "low"),
  data.frame(cv = coefficient_of_variation(case7), sd = "medium"),
  data.frame(cv = coefficient_of_variation(case8), sd = "high")) |>
  mutate(sd = factor(sd, levels = c("none","low", "medium", "high")))
```

```
mycolors <- c("#faf5f6","#e0abb4","#ad3b50","#5e0112")
p3 <- ggplot(plasticity_cv, aes(x = cv, fill = sd)) +
  geom_histogram(binwidth = 0.13, color = "black", alpha = 0.6) +
  geom_vline(
```

```
    data = plasticity_cv |> group_by(sd) |> summarise(mean_cv = mean(cv)),
    aes(xintercept = mean_cv),
    color = "black", linetype = "dashed", linewidth = 1
  ) +
  labs(x = "Coefficient of Variation", y = "Counts of simulations") +
  theme_minimal() +
  coord_flip() +
  xlim(0,7)+
  scale_y_continuous(breaks = c(30, 60))+
  facet_wrap(~sd, ncol = 4, strip.position = "top") +
  scale_fill_manual(values = mycolors) +
  ggtitle("Full Plasticity Scenario") +
  theme(
    legend.position = "none",
    axis.title.x = element_text(size = 18),
    axis.title.y = element_text(size = 18),
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14),
    plot.title = element_text(size = 20, face = "bold", hjust=0.5),
    strip.text = element_text(size = 16)
  )
p3
```

```
Warning: Removed 8 rows containing missing values or values outside the scale range
(`geom_bar()`).
```




```
ggsave("plot3.png", plot = p3, width = 5, height = 6, dpi = 300)
```

```
Warning: Removed 8 rows containing missing values or values outside the scale range
(`geom_bar()`).
```

```
plasticity_cv_table<- plasticity_cv |>
       group_by(sd) |>
            summarise(mean_cv = mean(cv))
plasticity_cv_table
```

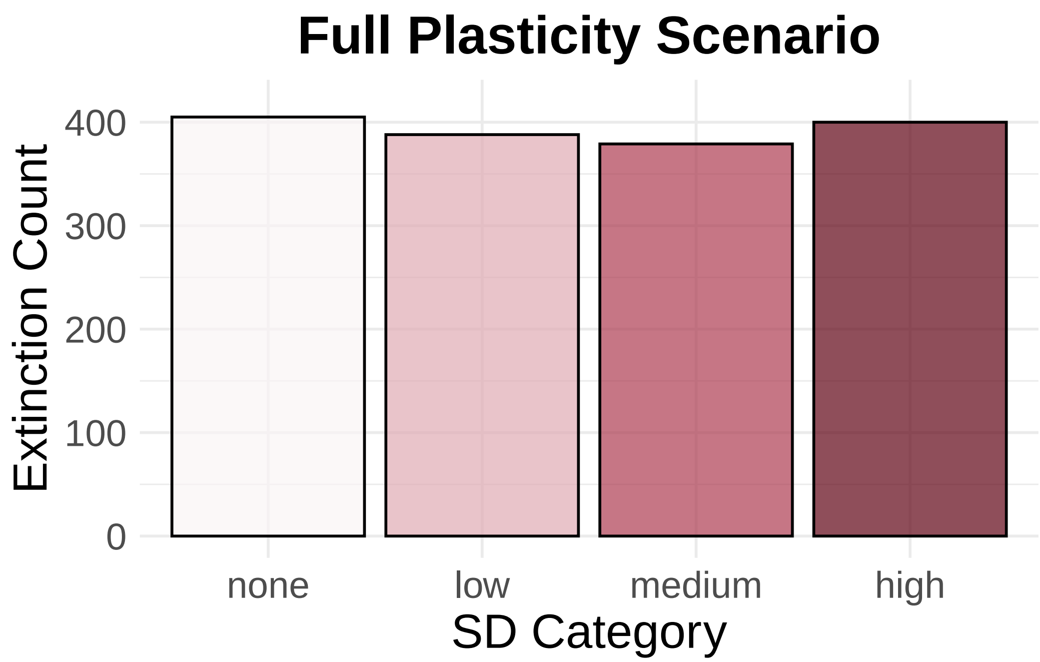
```
# A tibble: 4 x 2
  sd       mean_cv
  <fct>      <dbl>
1 none        1.63
2 low         1.64
3 medium      1.68
4 high        1.84
```

```
plasticity_extinction <- c(extinct_count(case5),extinct_count(case6),extinct_count(case7), ex
sd_labels <- c("none","low", "medium", "high")
```

```r
plasticity_extinction_df <- data.frame(
  count = plasticity_extinction,
  sd = factor(sd_labels, levels = sd_labels)
)
```

```r
p4 <- ggplot(plasticity_extinction_df, aes(x = sd, y = count, fill = sd)) +
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  scale_fill_manual(values = mycolors) +
  labs(
    x = "SD Category",
    y = "Extinction Count",
    title = "Full Plasticity Scenario"
  ) +
  ylim(0,420)+
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.title.x = element_text(size = 18),
    axis.title.y = element_text(size = 18),
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14),
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5)
  )

p4
```




```
ggsave("plot4.png", plot = p4, width = 5, height = 6, dpi = 300) #maybe put it back to 6 and
```

The count of extinctions is hugely greater than the trait given only at birth scenario, why???

```
data <- data.frame(
  SD = c("None", "Low", "Medium", "High"),
  `Mean CV Immutable`       = immutable_cv_table$mean_cv,
  `Mean CV Flexible`        = plasticity_cv_table$mean_cv,
  `Extinction % Immutable` = immutable_extinction_df$count /500,
  `Extinction % Flexible`  = plasticity_extinction_df$count /500
)
```

Sd	Mean CV immutable	Mean CV Flexible	Extinct % Immutable	Extinct % Flexible
None	1.123980	1.630573	0.216	0.810
Low	1.182187	1.639052	0.312	0.776
Medium	1.251113	1.679113	0.358	0.758
High	1.439831	1.839916	0.484	0.800